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Method Proposals Under a Shared Design Budget

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Abstract

Does proposing a design method improve battery-free service when proposal and design share one budget? We compare a fixed method, ordinary reflection, and a structured dependency and evidence-scope procedure on four fixed digital tasks, with three formal repetitions per task. All three conditions pass 12 of 12 complete-service evaluations. They use 64, 117, and 127 provider requests, respectively, including proposals and fresh design decisions. The registered selection retains the unchanged base method. Dependency and scope controls conservatively expand to complete feedback and establish no local-recomputation speed benefit. Four formal groups select delivery across a reset, but all reuse one historical ideal-model response; they are fresh Agent choices, not four new physical experiments or a new memory mechanism. The proposed methods are task-local instantiations. No global method transfer, change to the improver, recursive self-improvement, or hardware qualification is established.

1 Question and experimental object

The scientific question is whether a proposed design procedure adds service success once its own construction cost is included. Correctly classifying an edit or passing a tool contract does not answer that question. We separate a device program D, a method H that chooses and checks D, and an improver G that proposes and selects methods. A different successful D is not automatically a better H; generating a task-specific H does not establish an improved G. The present study evaluates method conditions while retaining negative and no-update outcomes.

The apparatus is a fixed ideal battery-free model with source arrivals, storage, state-dependent controller actions, event capture and a receiver. Three task IDs are `b-pulse-fast`, `b-volatile-reset`, and `b-latched-reset`. The fourth, `clite-query-window-derived-v1`, is a synthetic DC single-query derivative. The legal capacitance and transmit-floor ranges and service contract are fixed for each task. They cannot be edited by a method. These are four fixed digital tasks, not four physical devices or four independent environment families. The single-query mapping does not establish nonce security or multiple-query operation. Physical and RF qualification remain UNKNOWN.

The paper's main comparison uses three of the eight conditions. Its fixed-H cell is the same observed `complete-raw` cell used by the interface comparison; it is counted once. Likewise, `rich-static` also supplies the E4 JSON cell, while E4 table is a separate condition. The method subset does not create additional baseline groups or provider episodes. Table 1 counts the complete matrix once, and the main result below uses 36 of its 96 formal groups.

Stage	Tasks	Repeats/task	Groups	Episodes	Requests
Pilot	1	2	16	20	112
Formal	4	3	96	120	659
Total (distinct tasks)	4	—	112	140	771

Table 1: The full registered eight-condition matrix. A group is one task/condition/repetition, including a proposal episode when present. The pilot task is already among the four formal tasks; counts of tasks are not additive.

2 Conditions and prospective comparison

All three method conditions use the complete-raw information setting. Every supplied record and the full fixed model definition are available; complete raw physical traces are retained by the host for audit. Task, source and tool access are matched within a task/repetition. The fixed-H condition requests one complete design attempt. Ordinary reflection first reviews the supplied observations and proposes a repeatable procedural revision, or no update. The proposed procedure additionally asks for a typed D/H distinction, dependency-correct evidence scope, explicit event order, and a distinction between necessary constraints and sufficient design heuristics. It supplies no winning device values. Both proposing conditions must produce an actual model-authored hypothesis, falsifier and procedural method text, or abstain.

Every group has the same ceiling of 24 provider requests, 1,200 seconds and two distinct D candidates. A proposal subcap of eight requests and 400 seconds is included in that total. Acquisition, failed actions, the proposal, fresh design and confirmation all count. Using the entire proposal subcap leaves at most 16 requests and 800 seconds for the rest of the group. Equality here means equal registered ceilings and accounting scope, not equal realized spending. A no-update outcome keeps its proposal costs and uses the base H for any admitted fresh design.

A changed H is consumed at a fresh task entry with a new decision context. An unchanged lower-level physical response may still be reused under exact consumed-input identity. These are distinct operations. Every arm begins with the same immutable cache manifest: 12 phase2 exact-input responses plus one registered derived baseline. Job overlays are isolated; later physical deduplication does not expose another arm's result before its query or enlarge its two-design allowance. All 28 matched proposal groups, including pilot groups, have passing recorded information, tool, model-alias, cache and total budget parity checks. These checks establish comparison eligibility, not method benefit. The provider reports `deepseek-flash`; the weights are not independently pinned.

2.1 Pilot and fixed repetition count

The two pilots use one task and reverse condition order in the second repeat. The registered rule chooses two formal repetitions only if complete-service and design signatures are stable in every matched arm; otherwise it chooses three. Pilot signatures are not stable across all arms, so the repetition count is frozen once at three before the formal comparison. No original failure is replaced and there is no outcome-dependent stopping after this freeze. Formal task and group order are fixed by the protocol.

Pilot method condition	Service PASS/groups	Requests	Mean/group
Fixed H (complete raw)	2/2	10	5.000
Ordinary reflection	2/2	22	11.000
Proposed procedure	2/2	21	10.500

Table 2: The six method-subset pilot groups, shown separately from the formal denominators. The full pilot has 16 groups and 112 requests. Both proposal conditions propose an update in each pilot group.

3 Formal result: no added service success at higher cost

Each method condition passes all 12 formal complete-service evaluations (Table 3). Across the four fixed task IDs, the success vector is $(3, 3, 3, 3)$ for fixed H, ordinary reflection and the proposed procedure. The 12 observations per condition are repetitions of four tasks. They are not 12 new tasks and do not support a general population success-rate claim. Complete service is determined independently under the same task-specific receiver, energy, transient and duplicate-handling criteria.

Condition	PASS/groups	H update/no-update	Requests	Mean	USD upper
Fixed H (complete raw)	12/12	—	64	5.333	0.4231806
Ordinary reflection	12/12	10 / 2	117	9.750	0.8190117
Proposed procedure	12/12	12 / 0	127	10.583	0.9367914

Table 3: Main formal comparison: four fixed tasks times three repetitions. Requests and cost include proposal and fresh D work. USD values are settled provider upper estimates, not invoices. A dash denotes no proposal stage.

Ordinary reflection proposes ten updates and returns no-update twice; the proposed procedure proposes in all 12 formal groups. The no-update groups stay in the denominator and retain their costs. The method subset uses 308 requests in total, including the 64 requests of the one shared fixed-H baseline. The proposing conditions use 117 and 127 requests versus 64 for the fixed method. Their means are 9.750 and 10.583 requests per group versus 5.333. Differences in device choice or intermediate reasoning do not create additional complete service successes.

The registered selection requires strictly more task-level complete-service success than both the fixed method and reflection, no lost task, and no higher mean request count. The proposed procedure has the same task-success vector and a higher mean request count. The decision is **KEEP_UNCHANGED_BASE_H**. It is an actual negative selection result under the registered objective. It does not show that reflection or structured reasoning can never be useful; this finite task set has a complete-service ceiling for the strong baseline.

3.1 Accounting and shared cells

Across the entire matrix, all 771 requests settle and no usage remains UNKNOWN. The 16 pilot groups use 112 requests and USD 0.7345845 upper; the 96 formal groups use 659 requests and USD 4.7132760. The combined provider upper estimate is USD 5.4478605. The matrix has 140 actual episodes: 112 design episodes and 28 proposal episodes, not 140 physical experiments. These whole-matrix totals also include the other interface conditions; they must not be charged again as a separate method batch. All proposal and fresh design costs are retained, including abstention and failure. There is no invoice audit. Codex assistance is unpriced and nonzero, so full end-to-end cost remains UNKNOWN.

4 Validity controls and event-order interpretation

Recomputation correctness and method benefit have separate evidential roles. The deterministic comparison annotates 107 existing admitted group responses; five groups have no admitted final response. Of the 107, 91 change D and 16 are exact unchanged-input controls. The four strategies in Table 4 are evaluated against those existing response identities, not four new paid trial arms. Additional scientific executions for this comparison are zero.

The coupled device affects controller actions and later physical state; an old trajectory cannot stand in for the changed load's service. For an H change, dependency, scope and full policies all require fresh Agent decisions from task entry. A physical exact-input cache hit does not restore validity to old Agent decisions. These are validity constraints and a null computational saving, not an observed method improvement.

Strategy	Recorded meaning for the coupled device change
Frozen trajectory	Invalid after changed input; retained as a negative control. Exact unchanged-input records remain positive controls.
Dependency recomputation	Conservatively expands to the same complete feedback response used by the full strategy.
Scope recomputation	Conservatively expands to that same complete feedback response; no valid suffix or smaller region is demonstrated.
Full feedback	Uses the actual complete response for the candidate and its consumed input identity.

Table 4: Correctness comparison on retained actual responses. The three valid strategies share one execution key within every admitted record. There is no local-recomputation speed benefit or extra scientific execution.

4.1 Four choices, one historical reset-retention response

The ordered audit finds four formal groups with a pending event delivered across reset: two fixed-H selections, one proposed-procedure selection, and one E4 table selection (Table 5). All select the same historical `b-latched-reset` response at 10 μF and a 45 ms transmit floor. Capture actions run from 3.7 through 7.9 ms, brownout occurs at 12.1 ms, and the same event is received at 46.9 ms. The audit binds event identity to the single-event ideal state machine. Physical qualification is UNKNOWN.

Condition	Formal selected groups	Repetition IDs
Fixed H / complete raw	2	2, 3
Proposed procedure	1	2
E4 table	1	3
All selected groups	4	One distinct response key

Table 5: Four fresh Agent choices, all reusing one historical physical execution key. The mechanism is the fixed model's ideal passive latch; no new memory mechanism or new physical experiment is demonstrated.

The remaining 83 successful formal groups deliver before any reset. The initial controls separately include a successful before-reset response, a volatile-latch response without valid delivery, and the historical latched cross-reset response. Relabeling these existing traces adds no experiment. A reset after delivery is not evidence that pending state survived reset; the legacy success-and-any-reset label is insufficient without event identity and capture–reset–delivery order.

5 Discussion and limits of the negative result

The apparatus makes two different questions observable. The validity controls show why changing a coupled input invalidates an old trajectory and why changing H invalidates its earlier decision span. The formal comparison then asks whether paying for an actual method proposal adds service success. The first question has a conservative correctness answer; the second has a negative result on the registered tasks. Passing an interface or hash check cannot substitute for an improvement in the second question.

The baseline already completes every formal task/repetition. The structured procedure does not create another success and costs more requests. This is a useful counterexample to treating more explicit reasoning or more submitted method updates as an automatic research gain. It does not identify which instruction caused a particular device choice, nor isolate the value of each reasoning step. No component ablation or significance test is claimed. Four fixed digital tasks and three repetitions provide a finite development comparison, not broad generalization over battery-free products.

5.1 Task-local methods and historical boundaries

The actual H texts are instantiated separately within each task/repetition. They are not one selected global H transported unchanged across new tasks. The proposing procedure G is not changed or compared through independent method lineages. Consequently, the result establishes neither global method transfer nor recursive self-improvement. More method text, a changed device, and a selected historical latch response are distinct objects and claims.

The prospective phase3 protocol does not erase phase2's failure. Its original nine-request proposal episode remains blocked at the host action/budget contract; recovering actual model-authored text was explicitly posthoc, and that phase's initial-cache parity was not established. Phase3 matches the initial cache and uses prospective role separation, but cannot retroactively make that earlier episode successful. Prior A/R fixed-space NO_GO, strict FIFO FAIL and product NOT_ACCEPTED also remain intact. Physical validation and product acceptance are separate from every service PASS reported here.

5.2 Frozen next step and reproducibility

The selection for the next phase retains the base H. Two frozen parameter derivatives, one with a stretched BF supply window and one with a C-Lite query at half power, are candidates for within-model digital generalization only. They have no participant exposure or scientific execution in the supplied freeze. They are not a new physical family or an executed transfer result. The historical `b-esr-fail` and `b-no-energy` controls have already been used and cannot be called unseen. The freeze's explicit historical-control classification takes precedence over the older holdout wording retained in the selection-rule string. A separately admitted execution would be required for the new tasks.

`phase3-binding.json` pins the seven primary S0 files, this manuscript's sources, the unchanged shared style and official English vector logo, and the compiled PDF. `render_phase3.py build` reads those files and recomputes only tables and counts before two TeX passes; it never runs the scientific apparatus, model, SDK smoke or tests. Small build, all-page visual and cleanup receipts are retained under `docs/research/phase3-20261020/methods/paper-final/`. The current writing adds zero provider requests, scientific executions or native calls. Earlier manuscripts, code, configurations, tests and handoffs are preserved. No fresh-clone or independent-host rebuild is claimed.